Physical Chemistry Cume Sept 25 2020

Quantum/Spectroscopy Announced Topic

1) What are the selection rules governing a) vibrational transitions and b) rotational transitions in molecules? (20 points)

2) Sketch a typical high resolution vibrational-rotational infrared spectrum of a gaseous diatomic molecule, and assign the peaks to the specific vibrational-rotational transitions. You may find it helpful to draw an energy level diagram of the vibrational and rotational energy levels of a diatomic molecule in terms of the vibrational and rotational quantum numbers to help you sketch the spectrum. (25 points)

3) Vibrational-Rotational transitions in the high-resolution infrared spectrum of gaseous $^{1}H–^{19}F$ molecules were observed at 2886.50 cm^{-1}, 2908.51 cm^{-1}, 2930.43 cm^{-1}, 2974.55 cm^{-1}, 2996.57 cm^{-1}, 3018.58 cm^{-1}.
   The mass of ^{1}H = 1.0079 a.m.u and the mass of ^{19}F is 18.9984 a.m.u,
   1 a.m.u = 1.6605×10^{-27}kg
   A) By considering the separation between pairs of lines, assign the transitions of the P and R branches of the spectrum. (15 points)
   B) Determine the bond length of $^{1}H–^{19}F$, assuming it behaves as a rigid rotor. (40 points)

You may find the following equations useful for a rigid rotor model:

$$E = hc\tilde{v}$$

$$E_J = BhJ(J+1) = hc\tilde{B}J(J+1)$$

$$B = \frac{\hbar}{4\pi I} = \frac{h}{8\pi^2 cI}$$

$$I = \mu R^2$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$